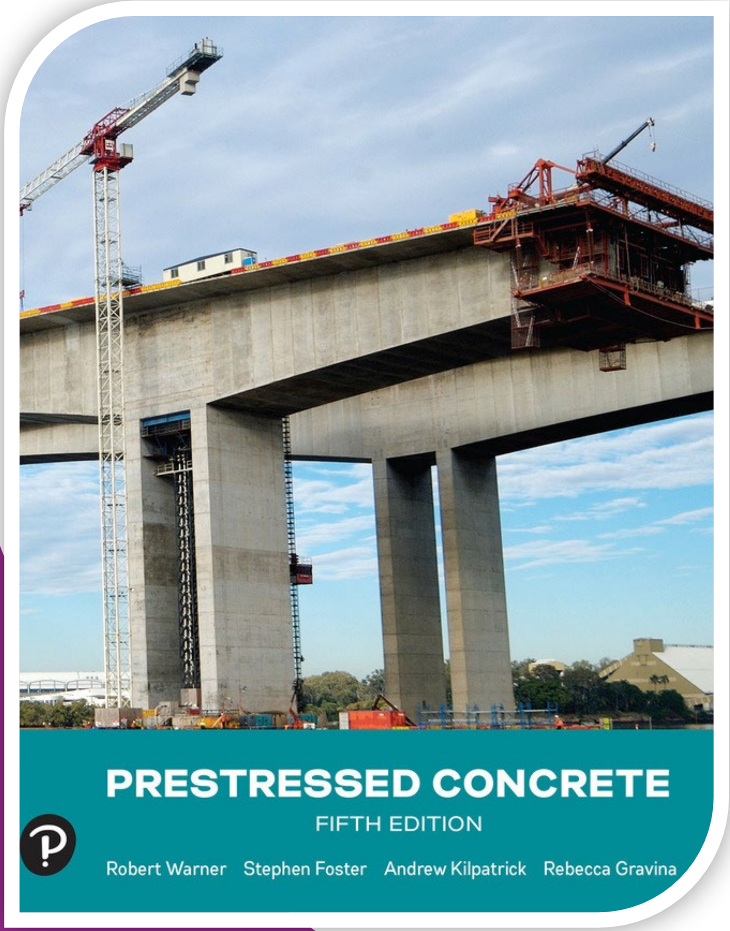

Introduction to Prestressed Concrete Structures

Week 1

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Please email for appointments

[Prestressed Concrete, Pearson Original Edition | ISBN 9780655706397 | Concrete | Civil Engineering | Engineering | Store]

Contents

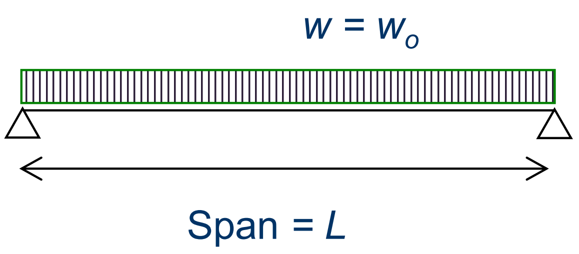
- Introduction to prestressed concrete
- Method of prestressing
- Member behaviour and stress distribution
- Line of pressure (line of action)
- Straight, deflected and curved tendons
- Loss of prestress
- Examples and exercises

INTRODUCTION TO PRESTRESSED CONCRETE

Introduction to Prestressed Concrete

What is prestressed concrete and why?

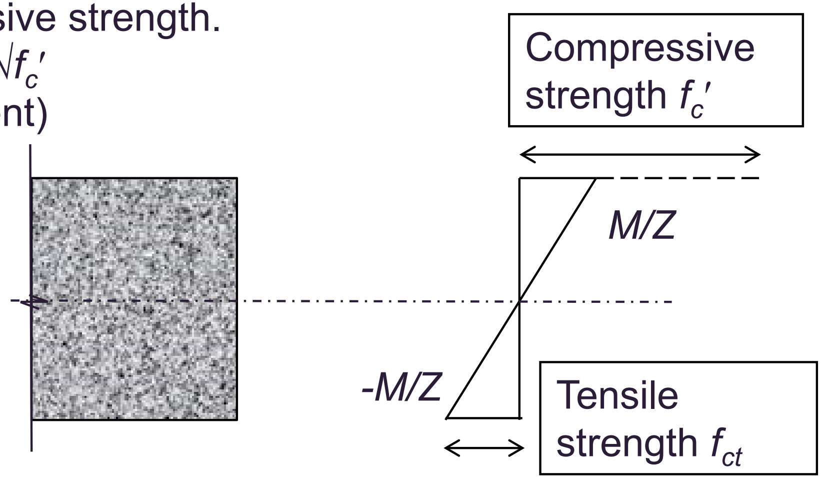
Simply-supported beam



If it is a plain concrete beam with rectangular cross section, the stress distribution is (symmetric stress distribution):

Note: Tensile strength of concrete is much lower than its compressive strength.
e.g. $f_{ct} \approx 0.6\sqrt{f_c'}$
(a few percent)

flexural concrete tensile strength



Z = elastic section modulus = I/y

Observation

The tensile strength of concrete will be reached quickly by increasing *w* and the compressive strength of concrete is considerably under-utilised.

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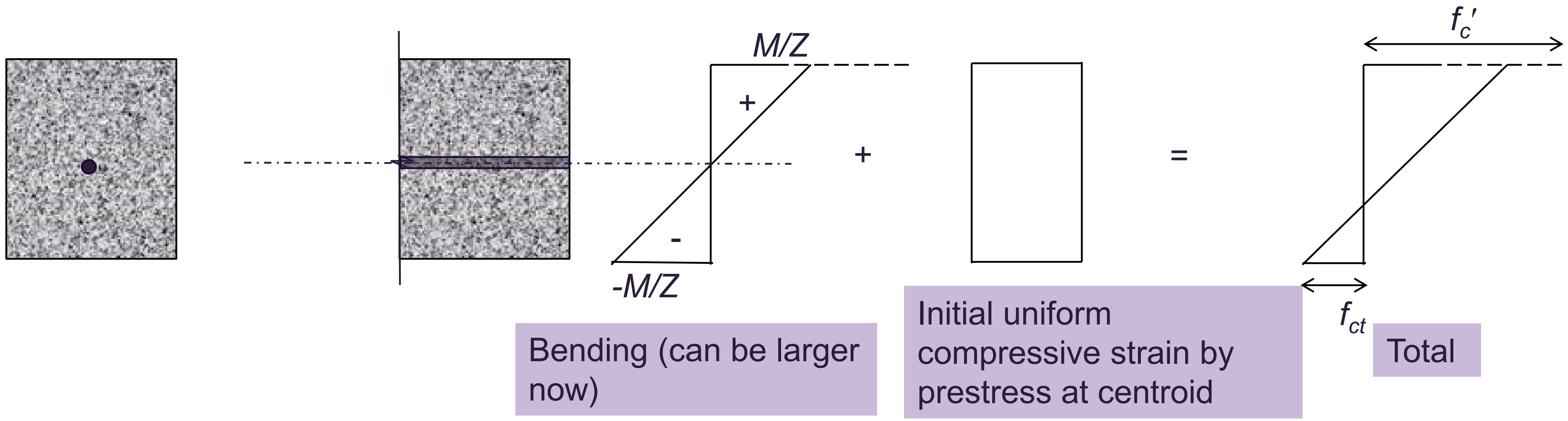
How to improve?

To fully ultilise the compressive strength of concrete and to postpone the bottom fibre from reaching the concrete tensile strength soon, we can impose an initial uniform compressive strain to the concrete.



Observation

By initially prestressing the section at centroid, a uniformly distributed initial compressive strain is introduced. The compressive strength of concrete is better utilised, and a heavier external load $w_1 > w$ can be sustained.

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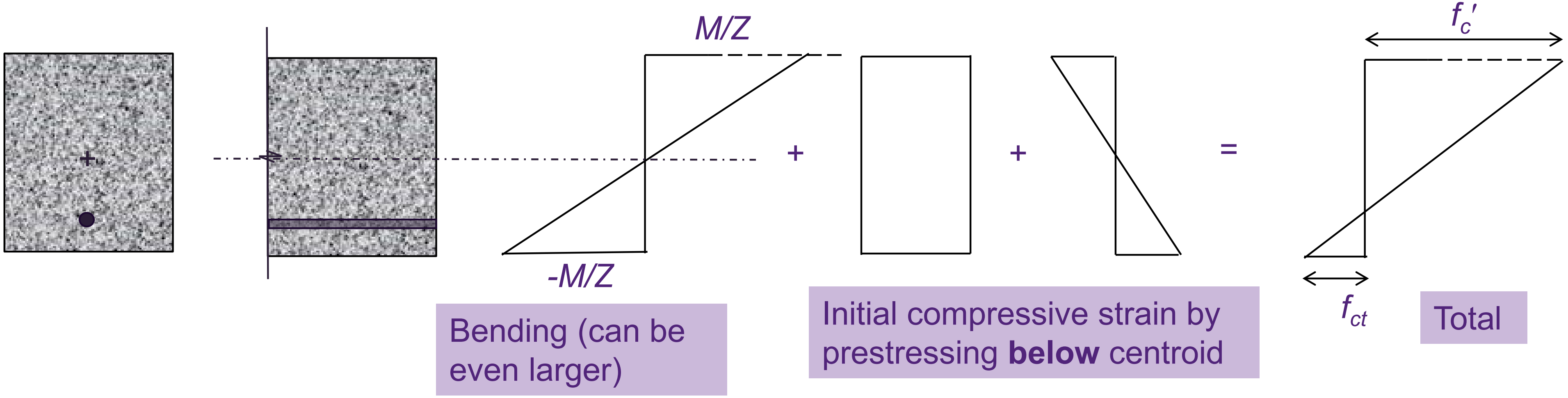
Further improvement possible?

Further, if an opposite strain distribution to that induced by bending can be applied simultaneously due to prestressing, an even heavier load w_2 can be sustained, where $w_2 > w_1 > w_o$.



Observation

By initially prestressing the section below centroid, an even heavier load can be sustained. Since concrete compressive strength >> tensile strength, the opposite strain distribution can delay the bottommost fibre from reaching its tensile strength.

Introduction to Prestressed Concrete

Prestressed concrete structures:

- Prestressed concrete structures an initial prestressing force (compression) is applied to the concrete section by stretching the prestressing steel. Hence, the section is subjected to an initial compressive strain even before any external load is applied.

- In ordinary plain concrete section (i.e. no prestress), the section normally would fail when the concrete tensile strength is reached.

- By applying the prestress at the centroid of the section, a uniformly distributed compressive strain is induced. The load-carrying capacity increases.

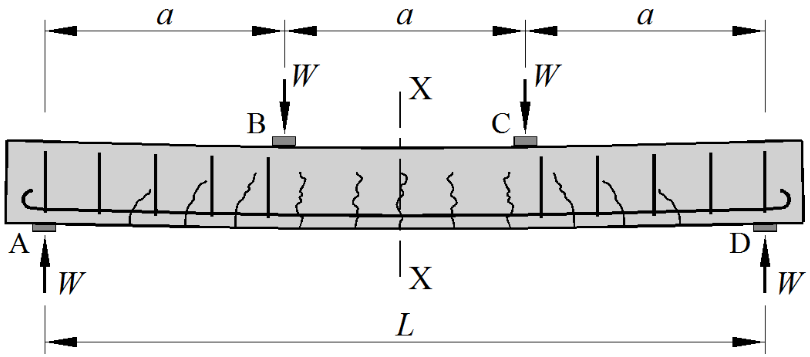
Introduction to Prestressed Concrete

- Since normally the concrete tensile strength will govern the design ($f_{ct} << f_c'$), the load-carrying capacity can further be increased by applying some more compressive stress at the bottom concrete fibre before loading.

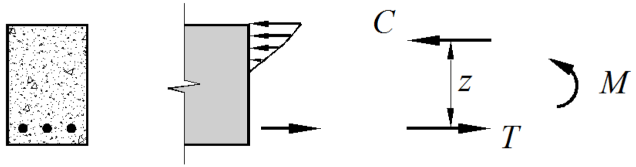
- This can be achieved by applying the prestressing force below the centroid of the section. Thus tensile strain and compressive strain will be induced at the top and bottom concrete fibres respectively. The beam is subjected to hogging moment after **transfer**.

- Note if the prestressing force is too large, tensile stress may occur at the top of section, which should be avoided (should within tensile strength of concrete).

Introduction – Reinforced concrete beam with external loads



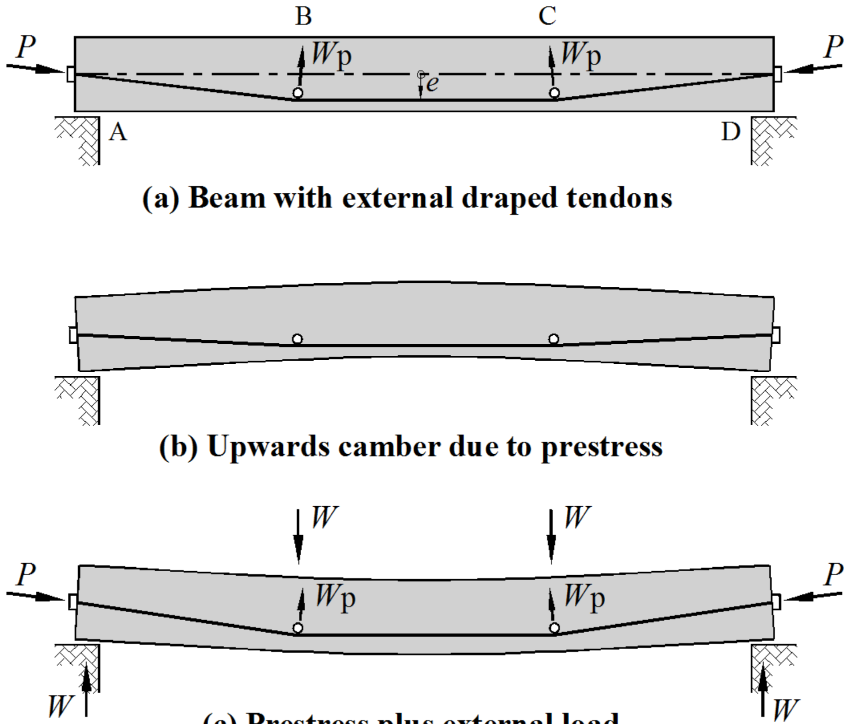
(a) Reinforced concrete beam under service load



(b) Stresses and internal forces at section X-X

Introduction – Prestressed Concrete

Introducing prestress into a concrete flexural member creates system of permanent compressive stresses in regions where tensile stresses will subsequently be induced by external service loads



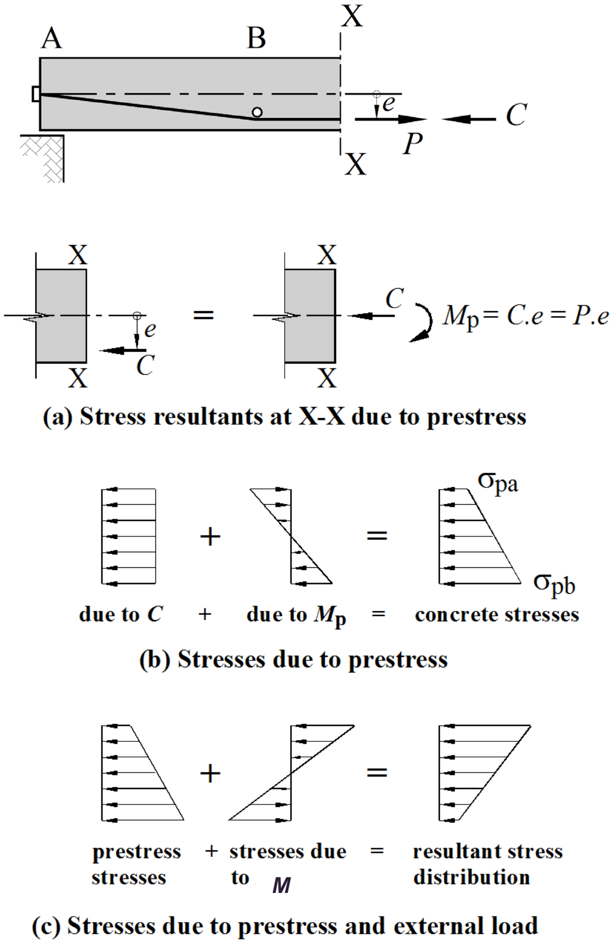
(a) Beam with external draped tendons

(b) Upwards camber due to prestress

(c) Prestress plus external load

Introduction – Equilibrating forces at section X-X

Stresses in the concrete induced by the prestress



Introduction to Prestressed Concrete

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History of prestressed concrete structures

- Prestressed concrete is the most recent of the major form of construction to be introduced into structural engineering.

- In early days, adopting of low-strength steel made the idea of prestressing not practically feasible because the long term loss of prestress due to creep and shrinkage is too significant.

- Until early 20th century, high-strength steel (>1,500 MPa) was applied successfully in prestressed concrete structures. Because of high-strength steel, the relative long term loss is smaller and hence the advantage of using prestress is justified.

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Example of prestressing technique (1)

- Barrels (for wine storage) made from separate woods and were kept together by metal hoops.
- The metal hoops are smaller in diameter than the diameter of the barrel, and are forced into place over the staves, so tightening them together and forming a watertight barrel.


Introduction to Prestressed Concrete

<u>Example of prestressing technique (2)</u>

- Cartwheels were similarly prestressed by passing a heated iron tyre (initially smaller in diameter) around the wooden rim of the wheel.

- On cooling, the tyre would contract and be held firmly in place on the rim, thus strengthening the joints between the spokes and the rim by putting them into compression.

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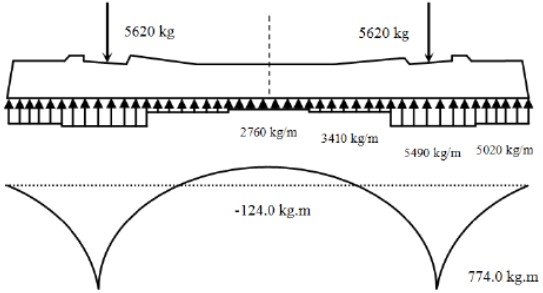
Example of prestressing technique (3)

- Railway sleepers
- Vibration – tensile stress; hogging moment – tensile stress on top ➔ cracking of concrete
- Prestressed sleeper to get rid of tensile stress in sleeper under load



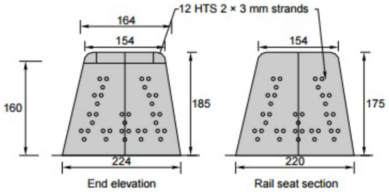


Fig. 7.14 PCS-17 concrete sleeper for MG (units in mm)

Invention of Prestressed concrete

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History…

Used high tensile steel wires, with ultimate strength as high as 1725 MPa and yield stress over 1240 MPa. In 1939, he developed conical wedges for end anchorages for post-tensioning and developed double-acting jacks. He is often referred to as the **Father of Prestressed concrete.**

Eugene Freyssinet
(France)

1938 Hoyer, E., (Germany)
Developed 'long line' pre-tensioning method.

1940 Magnel, G., (Belgium)
Developed an anchoring system for post-tensioning, using flat wedges.

J.D. COLLEGE OF ENGINEERING, NAGPUR

Post tensioned concrete

https://www.youtube.com/watch?v=z_67Jkz9Tz0

Prestressed concrete
https://www.youtube.com/watch?v=pjwrXLWhISE

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<u>Application (Civil structures)</u>

- Bridge engineering – for construction of long-span concrete bridges.
- Segmental construction – not discussed in this course
- Beam-and-slab deck construction: Comprise precast prestressed bridge segments lifted into position. Then in-situ slab to be cast on top.
- This is very effective in limiting the tensile stress at the bottom concrete fibre. Particularly useful for long-span structures as the moment due to dead load is quite significant – e.g. in lifting/transporting.
- One of the most popular bridge construction methods in Queensland

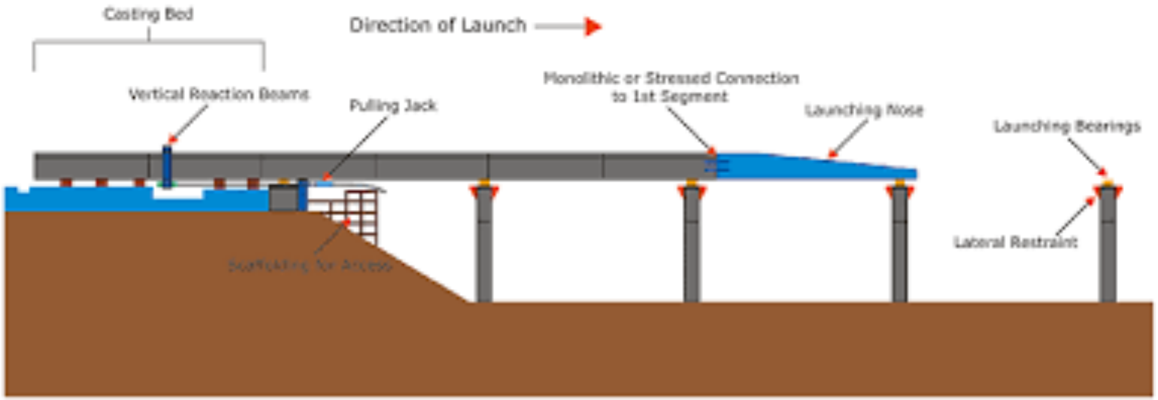
Segmental construction

- Balanced cantilever

- Progressive span-by-span segmental construction

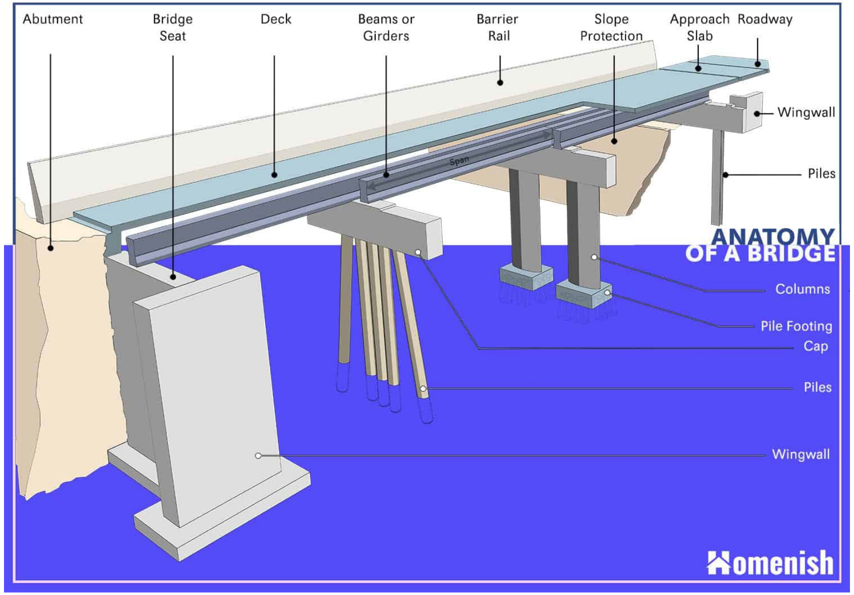
Segmental construction

- Incremental launching



Beam-and-slab deck construction

- Precast prestressed beam + cast in-situ slab

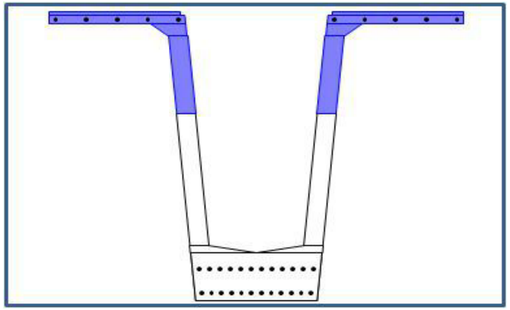


Introduction to Prestressed Concrete

- Popularly used beam section is called “Super-Tee” section. However, it is not in T-shape, but in U-shape.

- Normally the beam section is precast and prestressed in the precast yard and then lifted into position on site. Then, in-situ concrete will be cast on top of the beam, which will be the deck of bridge.

- Apart from Super-Tee, other common beam section include I-section, hollow (“box”) section, … etc.

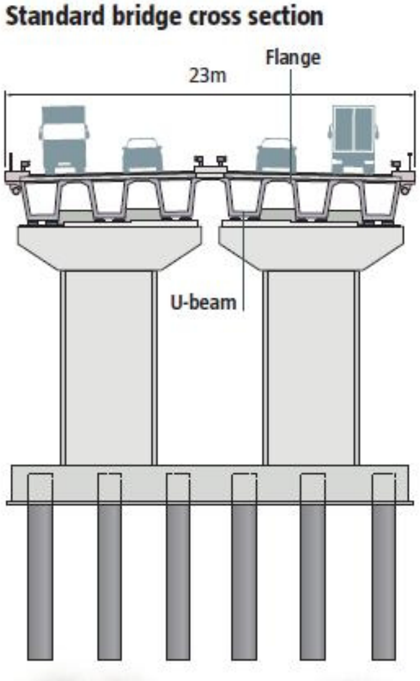


Introduction to Prestressed Concrete

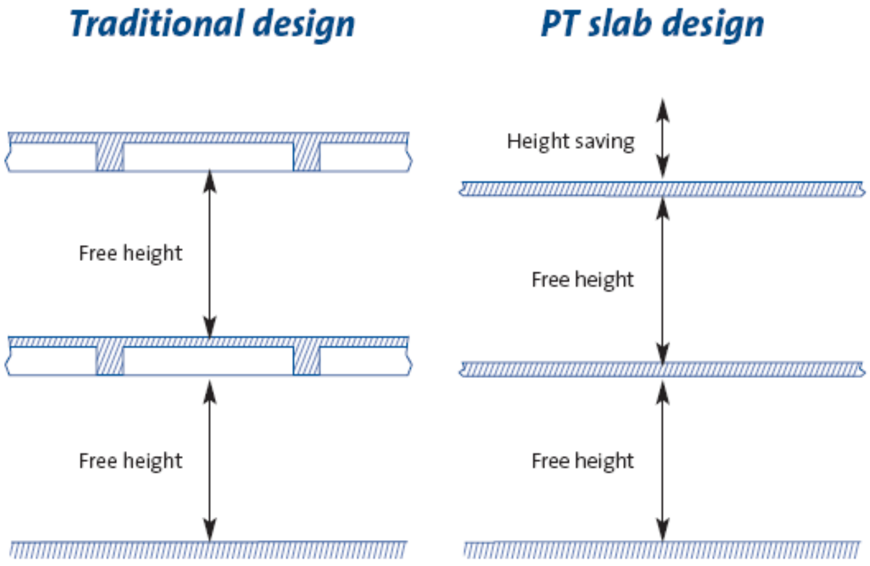
<u>Application (Building structures / water retaining structures)</u>

- Mostly in the form of simply supported precast floor and roof beams.

- Example:
  - In-situ prestressed concrete beams
  - In-situ prestressed concrete flat slab



Suspended floor systems

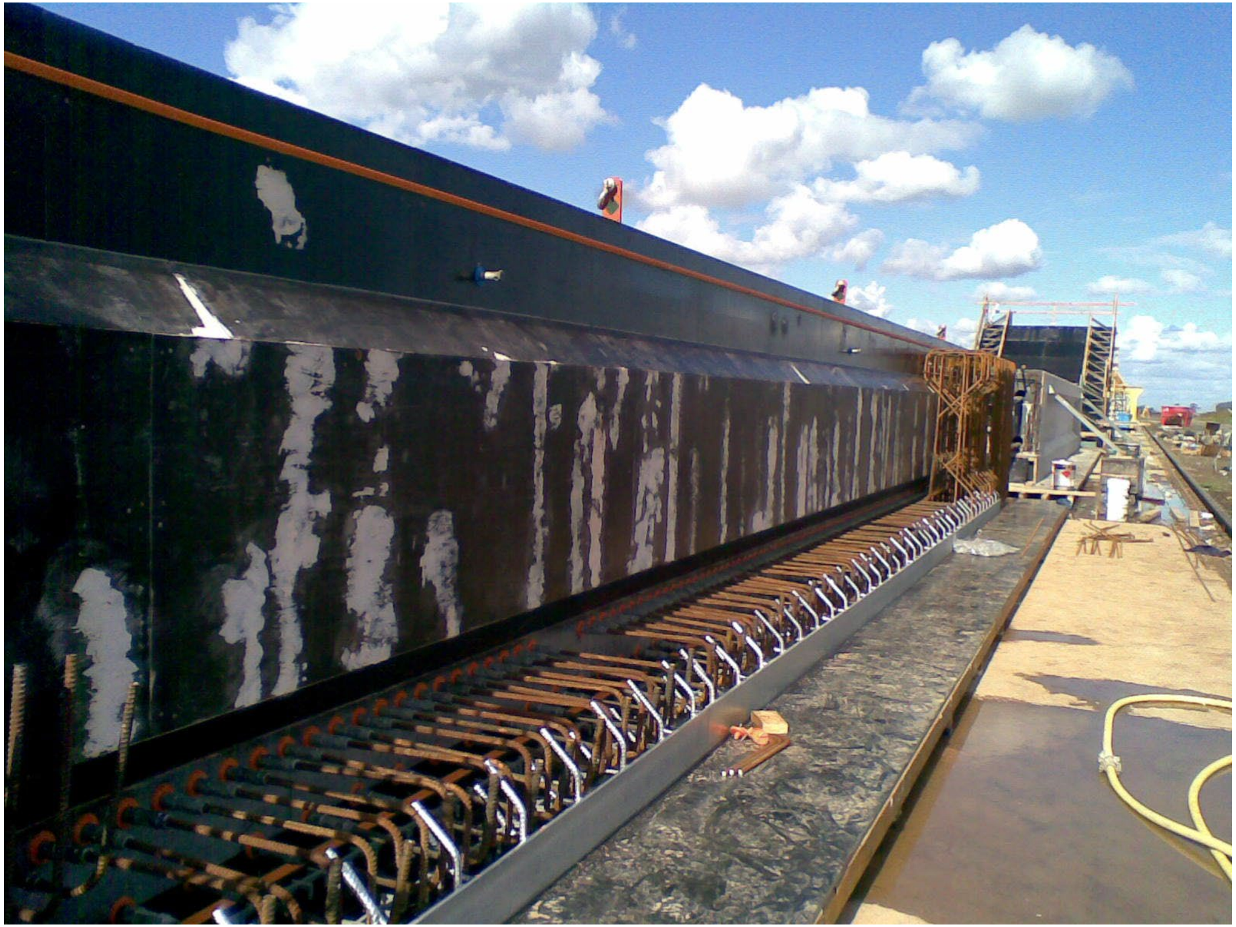


Prestressed Concrete piles

Prestressed I-beam

Note cross section & provision for starter bars! This is a railway bridge

Formworks, reo, Reidbars, cast-in angle



Formworks, reo, Reidbars, cast-in angle

Post-tensioning duct

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Post-tensioning duct – Note end block!

Post-tensioning duct, cast-in angle

Post-tensioning operation

Beams being delivered to site in specialised trucks

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Prestressed Planks- note strands epoxied

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Super T Beams- mould & hydraulic Jack

Super T Beams- Lifting monocranes

Super T Beams- strand coils 15.2mm

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Super T Beams- Reinforcements

Super T Beams- reo cage, strand plate, stopend

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Super T Beams- Edgeform & styrene

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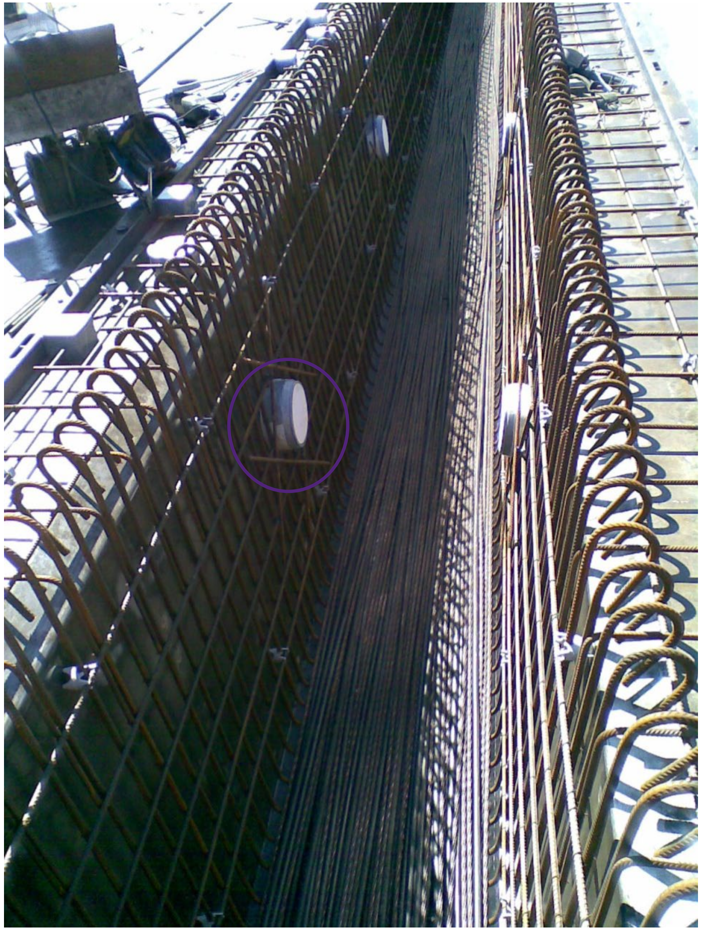

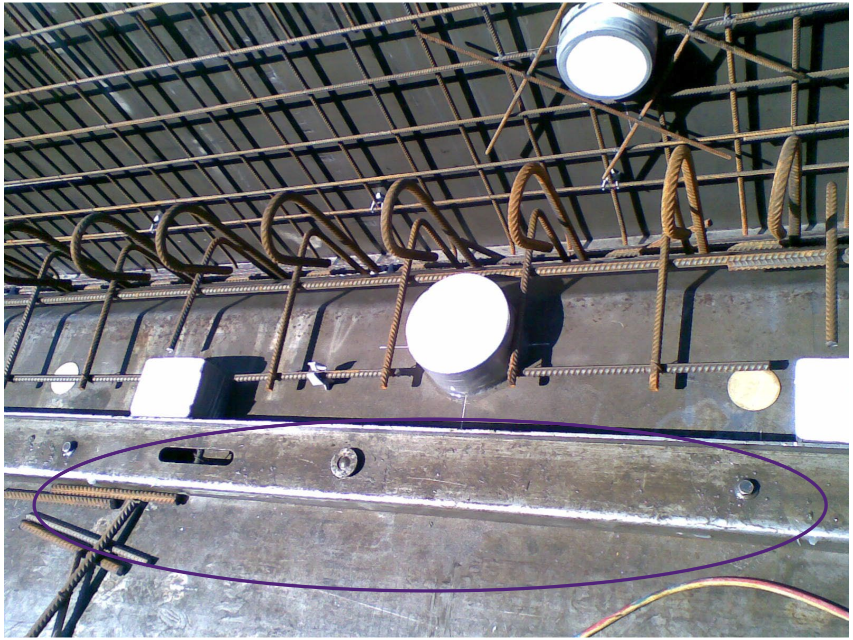

Super T beams – Cover & Boiler

Super T beams – Voids, web stiffeners, lifters, ligs

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Super T beams – transport to site!

Super T beams – installation on site with crawler cranes

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Super T beams – Bearing pad

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Introduction to Prestressed Concrete

<u>Advantages of PC structures:</u>

- For a given span and loading, a smaller section of prestressed concrete member is required. It increases effectively the load-carrying capacity to weight ratio (esp. Long-span).

- Prestressing force can be selected such that the concrete member can be designed as crack free during servicing stage, i.e. no tensile stress in the cross section.

- Downward deflection can be controlled. Because prestressing force will cause upward deflection (i.e. camber) to the beam and so the resulting downward deflection at servicing stage due to all service loads will then be smaller. (c.f. no initial upward deflection in RC structures – all deflections are downward)

Introduction to Prestressed Concrete

Disadvantages of PC structures:

- Hogging moment during the transfer stage, hence tensile stress at transfer of prestress is possible and should be limited to within concrete tensile strength (immature).

- That is, both minimum and maximum loading cases should be checked in PC structures (c.f. only maximum load in RC structures).

- Short- and long-term losses of prestress. Thus, high-strength steel should be adopted such that the relative loss can be minimised.

- Extra design considerations: Anchorage for prestress; maximum and minimum load conditions; large jacking force, safety … etc.

- Long term creep effect is higher in PC than in RC as the compressive stress is larger.

Introduction to Prestressed Concrete

Why prestress ?

Improve service-load behaviour

Reduce cracking

- Excessive cracks are unsightly
- Cracks may cause durability problems
- Structure may be liquid retaining
- Cracks reduce the stiffness of the member

Reduce deflections

- Usually the depth required to satisfy deflection requirements in an PSC beam is about 70% of that of an RC Beam.

METHODS OF PRESTRESSING

Methods of Prestressing

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Pre-tensioning

- To stress the tendons before the concrete is cast. The tendon force is transmitted to the concrete through the bond.
- The steel tendons, in the form of wires or strands or cables, are tensioned between end anchorages and the concrete member cast around the tendons.

Wires

Strand (7-wire)

Strand (19-wire)

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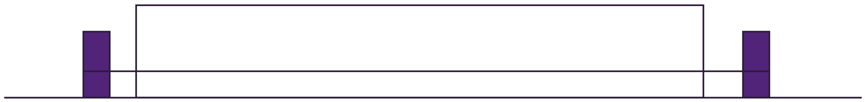
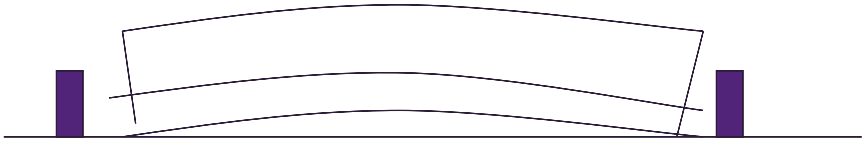
Methods of prestress

Pretensioning

Tendons stressed between abutments



Concrete cast and cured



Tendons released and prestress transferred

Methods of Prestressing

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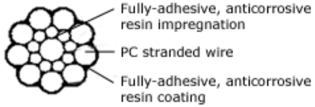
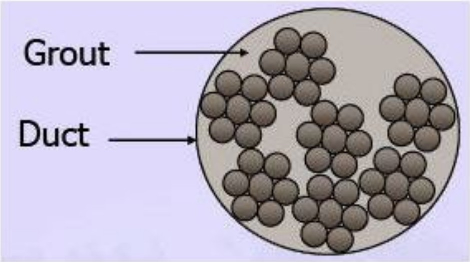


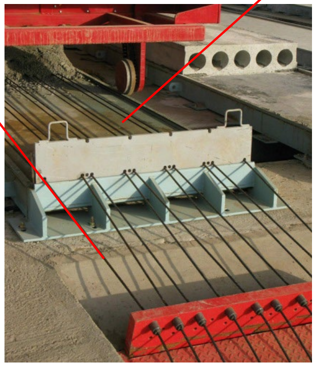
Cable



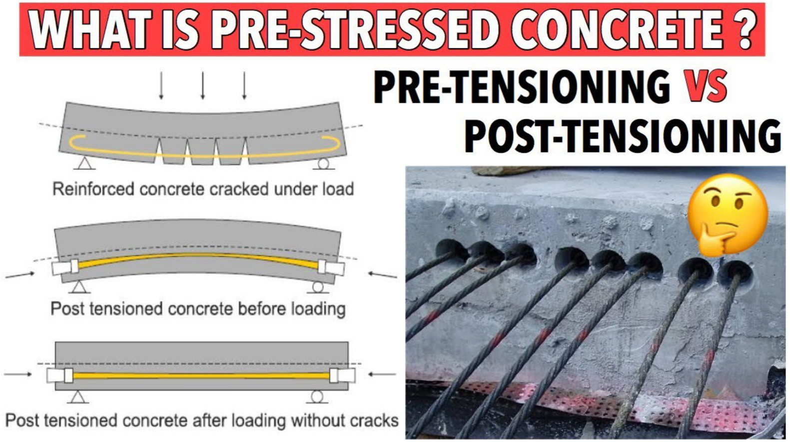
Strands

Prestressing bed

Stressing in progress



Methods of Prestressing



Strands as prestressing steel

Methods of Prestressing

- Once the concrete has gained sufficient initial strength, the end anchorages are released.

- The prestress force is transferred to the concrete through the **bond** between the concrete and the tendons.

- After stress transfer, the protruding ends of the tendons are then cut away to produce the finished concrete member.

- Ideal for factory production of large quantities of precast prestressed concrete members, e.g. Super-T beams. (Precasting bed should have no restraint against movement of members – otherwise there will be secondary moment)

Methods of Prestressing

- Normally for straight tendon, or segments of linear tendon profile with sharp change (“kink”).

- To avoid overstressing the member at ends, where the external bending moment is small, debonding of tendons at ends is required.

Methods of Prestressing

Post-tensioning

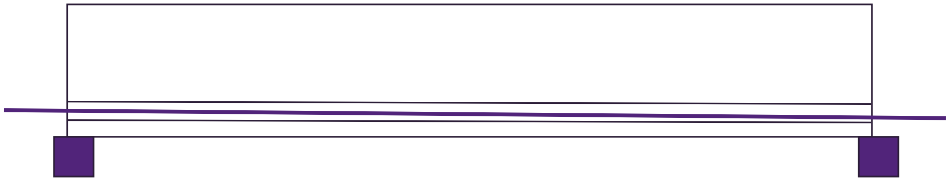
- Basically, this is to stress the tendons after the concrete is cast. The tendon force is transmitted to the concrete through the **anchorages** at ends of member.

- The prestress force is applied by jacking steel tendons against an already-cast concrete member.

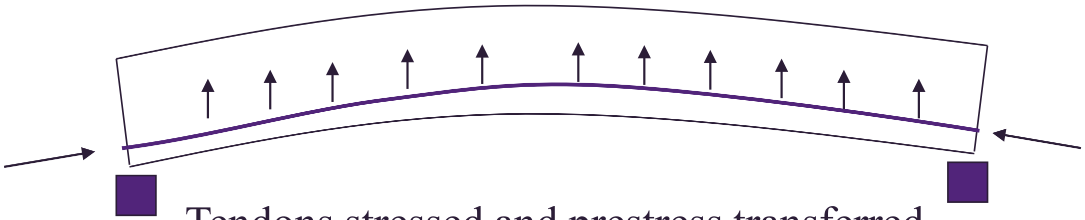
Post-tensioning stressing

Post-tensioning ducts

Post-Tensioning



Beam cast with an unbonded tendon inserted in a duct



Tendons stressed and prestress transferred

Tendons anchored and ducts grouted

It is normal for one end to be anchored against concrete (dead end) and the other end "jacked" against concrete. This end that is being jacked is known as the live end.

Methods of Prestressing

- Very common in in-situ PC members
- Ducts are cast in the concrete and then tendons are threaded through.
- Ducts can be inside the concrete (internal prestressing) or outside the concrete (external prestressing).
- The jacking force is transferred to the concrete members by special built-in anchorage (ends of members / also intermediate points).
- Unlike pre-tensioning, in post-tensioning the wires or strands are grouped into large tendon/cables and fixed to the same anchorage – save $$ as anchorage is expensive and induces loss of prestress.

Methods of Prestressing

- Heavily reinforcing cage providing confining pressure to avoid bursting is required in the proximity of the anchorages (none in pre-tensioning)

- Stressing can be applied in stages (sequentially) – not in pre-tensioning

- Capable of incorporating curved tendons

Non-linear tendon profile and internal prestressing (Bonded tendons)

External prestressing (Unbonded tendons)

Prestressed Concrete: Practical Examples

Post-tensioning

I beams (typically 20m)

Slabs

Large tanks (circumferentially prestressed)

Prestressing

Super T beam (typically 32m)

Planks (typically 10-12m)

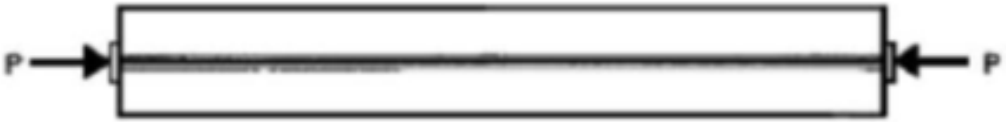
Concrete piles, bridge decks

MEMBER BEHAVIOUR
AND ELASTIC STRESS DISTRIBUTION

Member Behaviour and Stress Distribution

Member behaviour of PC simply-support beams

- Member behaviour will be discussed by considering the stress distribution of a prestressed member with rectangular cross section
- Different stages: At transfer and at service. (Both important)
- Consider a rectangular beam with *b* and *h* containing a strand producing a prestressing force *P* at centroid, i.e. *h*/2.



Simply-support beam with prestress applied at centroid

Member Behaviour and Stress Distribution

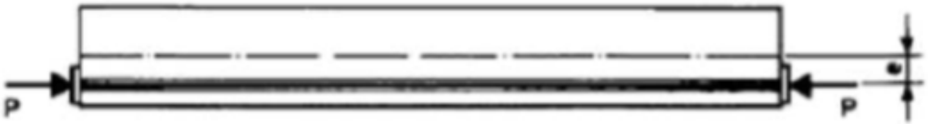
- Then, only direct compressive stress is produced, given by σ_c:

$$\sigma_c = \frac{P}{bh} \qquad \text{where } bh = A_c$$

- Sign convention: Compressive > 0

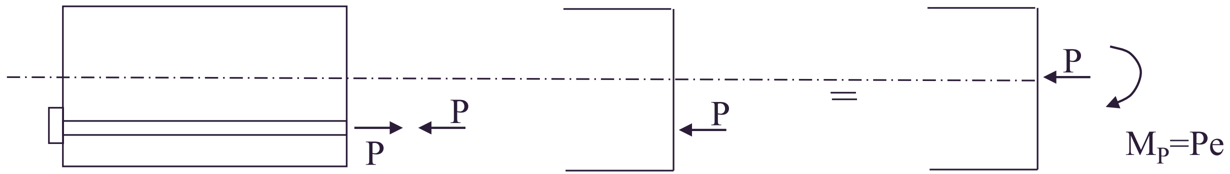
- If the strand is now at eccentricity *e* below the centroidal axis, then there is bending stress σ_P induced in addition to σ_c.

- In this case, there is hogging moment in the beam, i.e. tension at top and compression at bottom.



Simply-support beam with prestress applied at eccentricity, e

Internal force and moment in concrete due to prestress only



Tensile force in tendon, and equal compressive force in concrete cross-section

Axial force P and moment M_p in concrete section due to prestress

Member Behaviour and Stress Distribution

- The bending stress produced by the **strand** can be evaluated by the following formulas:

Top:

$$\sigma_{Pe,t} = -\frac{Pe}{Z_t}$$

Bottom:

$$\sigma_{Pe,b} = \frac{Pe}{Z_b}$$

- Z_t and Z_b are the elastic moduli of top and bottom fibres.

$$Z_t = \frac{I}{y_t} = \frac{bh^2}{6}$$

$$Z_b = \frac{I}{y_b} = \frac{bh^2}{6}$$

$$\text{where } y_b = y_t = \frac{h}{2} \text{ and } I = \frac{bh^3}{12}$$

I is the moment of inertia of the section

Member Behaviour and Stress Distribution

- The total stress due to prestressing at top and bottom surfaces of the beam can be obtained by adding the above stresses up:

Top:
$$\sigma_t = \sigma_c + \sigma_{Pe,t} = \frac{P}{bh} - \frac{Pe}{Z_t}$$

Bottom:
$$\sigma_b = \sigma_c + \sigma_{Pe,b} = \frac{P}{bh} + \frac{Pe}{Z_b}$$

- At transfer (i.e. the instant when the prestressing force is transfer from the strand to the concrete), the beam is only carrying self-weight. Suppose the self-weight creates bending **moment *M_{SW}***, the stresses are:

Top:
$$\sigma_{B,t} = \frac{M_{SW}}{Z_t}$$

Bottom:
$$\sigma_{B,b} = -\frac{M_{SW}}{Z_b}$$

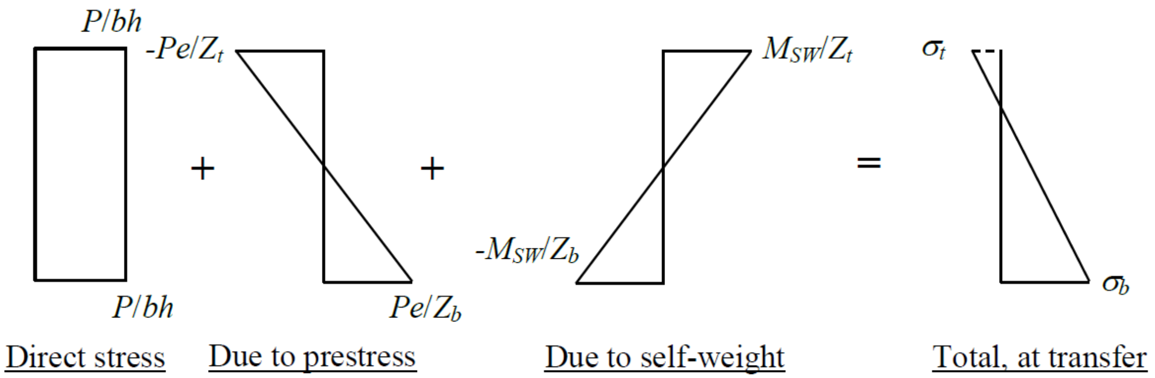
Member Behaviour and Stress Distribution

- The total stress at transfer will be:

Top
$$\sigma_t = \sigma_c + \sigma_{P,t} + \sigma_{B,t} = \frac{P}{bh} - \frac{Pe}{Z_t} + \frac{M_{SW}}{Z_t}$$

Bottom:
$$\sigma_b = \sigma_c + \sigma_{P,b} + \sigma_{B,b} = \frac{P}{bh} + \frac{Pe}{Z_b} - \frac{M_{SW}}{Z_b}$$

- Stress distribution at transfer:



(Note: Dotted line means that it can be compression or tension depending on the magnitude of each stress component)

Member Behaviour and Stress Distribution

- After transfer and the concrete has gained sufficient strength, additional external dead and live loads will act on the beam.

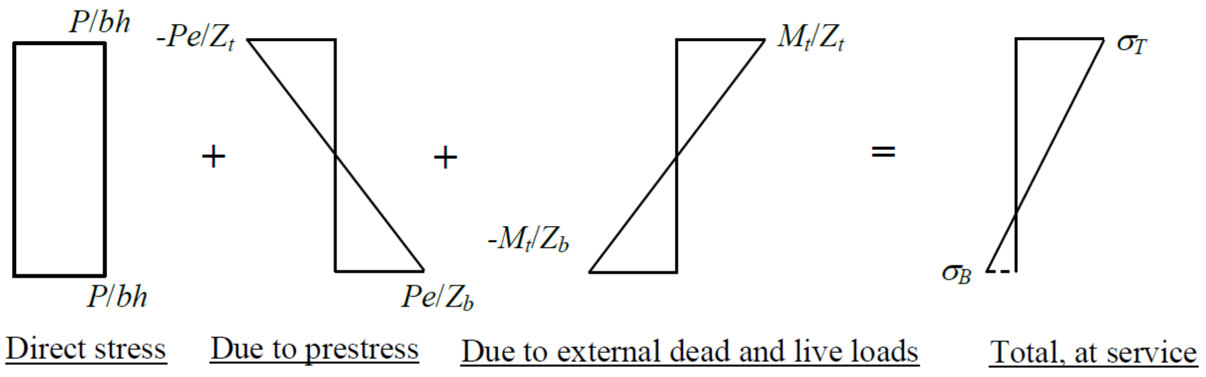
- The total bending moment is now M_t, which is the total moment including self-weight, dead and live loads during service. The stresses at top and bottom fibres are:

Top
$$\sigma_t = \sigma_D + \sigma_{Pe,t} + \sigma_{B,t} = \frac{P}{bh} - \frac{Pe}{Z_t} + \frac{M_t}{Z_t}$$

Bottom:
$$\sigma_b = \sigma_D + \sigma_{Pe,b} + \sigma_{B,b} = \frac{P}{bh} + \frac{Pe}{Z_b} - \frac{M_t}{Z_b}$$

Member Behaviour and Stress Distribution

- Stress distribution at service:



(Note: Dotted line means that it can be compression or tension depending on the magnitude of each stress component. Normally small tensile stress is expected).

Member Behaviour and Stress Distribution

<u>Design implications:</u>

- From the above stress distribution, we can have some design implications.

- In RC members, normally the top surface is in compressive stress whereas the bottom surface in tensile stress. Hence in design, the uppermost concrete fibre should be checked against the maximum allowable compressive stress, while the reinforcing steel should be checked against the maximum allowable tensile stress.

- Nevertheless in prestressed concrete, both compressive and tensile stresses could become critical on the top and bottom surface of the concrete.

Member Behaviour and Stress Distribution

- At transfer, the gravity load is the minimum and so does the compressive stress at the top. If the prestressing force or the eccentricity is large, tensile stress will occur at the top.

- The top fibre should then be checked against the maximum allowable design limit of (immature) concrete tensile strength.

- The bottom fibre should be checked against the maximum allowable design limit of (immature) concrete compressive strength.

- At service, the gravity load is the maximum (like RC member).

- The top fibre should be checked against the maximum allowable design limit of (mature) concrete compressive strength. The bottom fibre should be checked against the maximum allowable design limit of (mature) concrete tensile strength.

Member Behaviour and Stress Distribution

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- Then, by choosing a suitable pair of values of *P* and *e*, it is possible to completely eliminate the tensile stress at bottom concrete fibre at servicing stage ($\sigma_B \geq 0$). (Decompression)

- Nevertheless, the corresponding stress limits at transfer (minimum load) should be properly checked. (Not like in RC design)

- <u>At transfer:</u>
  Top: Maximum tension
  Bottom: Maximum compression
  <u>At service:</u>
  Top: Maximum compression
  Bottom: Maximum tension (No tension for non-cracked section)

Member Behaviour and Stress Distribution

Line of pressure (or line of action).

:

- Consider a rectangular beam with only one strand with eccentricity *e* (below centroid) subjected to external moment. Suppose there is no axial load, the neutral axis (n.a.) will divide the section into compression (above n.a.) and tension (below n.a.) zones.

- The tensile stress of concrete is usually ignored, and so there are two forces in the section. Concrete compression (above n.a.) and strand tension (below n.a.).

- By for equilibrium, we have:

$$\boxed{\text{Concrete compression} = \text{Strand tension}}$$

and then the n.a. depth can be solved.

Member Behaviour and Stress Distribution

- The location of the resultant concrete compression can be determined by considering the moment equilibrium condition.

- If the lever arm (i.e. the distance between the concrete resultant force and the tendon force) is z, the external moment is M and strand force is P, then we have:

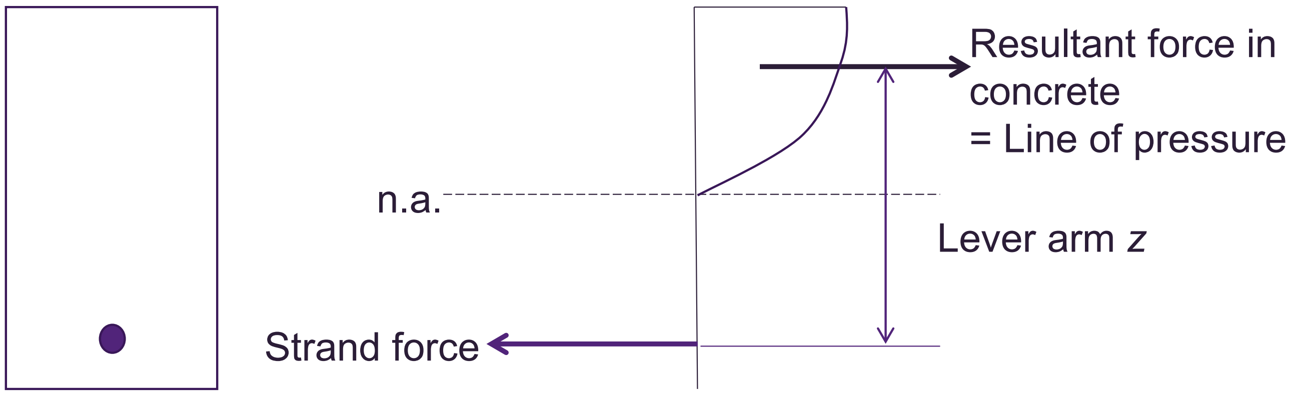
$$\boxed{M = Pz}$$

- For a UDL, M varies in a parabolic manner. Then, since P is a constant, z will also vary in parabolic manner along the span of the beam. (z follows the shape of BMD)

- It is defined that the locus of the resultant concrete force along the beam is the **line of pressure (or line of action)**.

Member Behaviour and Stress Distribution

- Forces within cross-section:



- It is clear that if the moment due to external load *M* is zero, then *z* is zero and the line of pressure is coincident with the strand – a profile called <u>**concordant profile**</u>.

DEFLECTED TENDONS

Deflected Tendons

Deflected tendons and internal induced forces

- When a straight tendon at eccentricity *e* is stressed in a prestressed concrete beam, at transfer, the beam will be subjected to a constant hogging moment due to prestressing, which is equal to *Pe*.

- In this case, only pure moment is induced to the concrete member without vertical and horizontal forces.

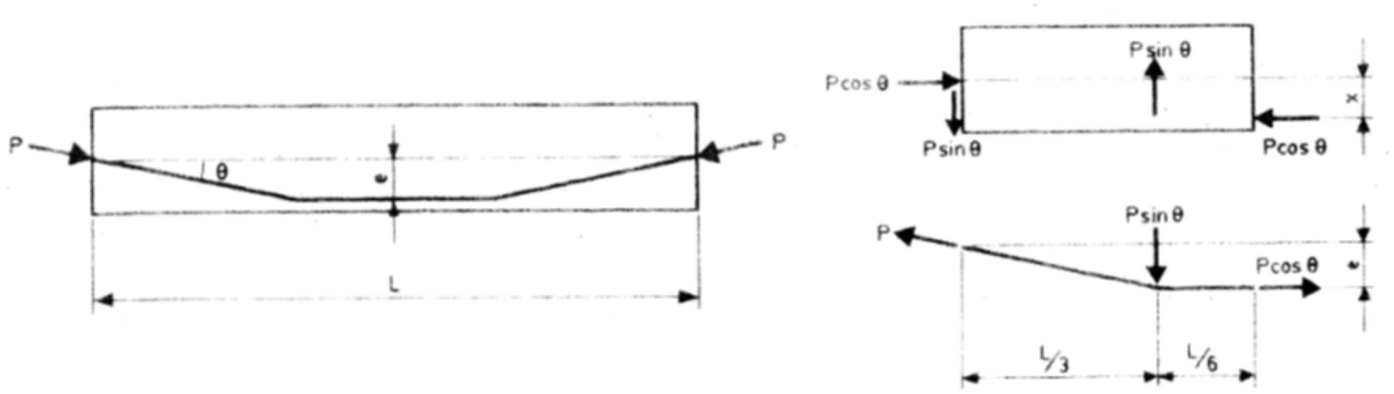
- Now consider a straight tendon but deflected at one-third points with eccentricity *e* and angle θ from horizontal.

- When applying *P* at both ends, concrete will be subjected to an upward force, whereas the tendon will be subjected to a downward force.

- Also horizontal forces will also be acting on the concrete and tendon at turning points.

Deflected Tendons

- The horizontal and vertical concrete forces can be determined by considering the free body diagrams below:

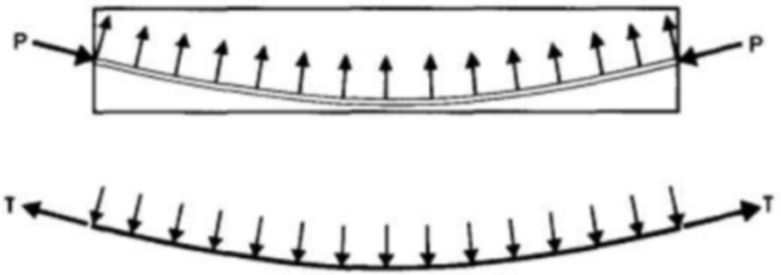


- Vertical force = P$\sin\theta$ (concrete upward);
  Horizontal force = P$\cos\theta$ (concrete compression)

- Therefore, when a tendon undergo a change of curvature (or angle), both horizontal and vertical forces are induced at that point.

Deflected Tendons

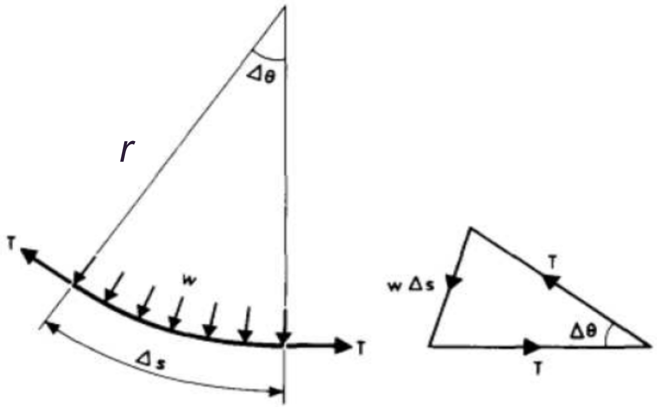
- For a curved tendon, because the curvature changes continuously along the tendon profile, vertical force will be created along the beam.



- Mathematically, the force can be evaluated by considering a very small segment of tendon length Δs:

Deflected Tendons

- Free body diagram considering a length of Δs in tendon:



- Since length is short, it can be regarded as straight. Then the total load on tendon is $w\Delta s$. Considering the force diagram on the right hand side, we have:

$$w\Delta s = 2T sin(\Delta\theta/2)$$

Deflected Tendons

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- Since Δs is small, $\Delta\theta$ is also small. Therefore, $\sin \Delta\theta \approx \Delta\theta$. Then the load *w*, which is the vertical force acting on concrete/tendon, can be evaluated as:

$$w = T\left(\frac{\Delta\theta}{\Delta s}\right) \approx T\left(\frac{d\theta}{ds}\right) = T\left(\frac{1}{r}\right)$$

- $d\theta/ds$ is the curvature of the curved tendon, which can be written as $1/r$ alternatively, where r = radius of curvature of the tendon.

Force on concrete = Prestressing force × curvature

LOSS OF PRESTRESS

Loss of Prestress

- It has been assumed that the prestressing force *P* is constant along the tendon length and does not change with time. However in reality, there are different types of losses that reduce *P*.

<u>Short-term loss</u>

- Elastic shortening – as the prestressing force is transferred to the concrete, the members shortens and the tendon shortens as well. The shortening of tendon will cause a decrease in the prestressing force.

- Pre-tensioned member: The loss should be estimated after transfer, which can be evaluated by the concrete stress at the level of tendon multiplied by the modular ratio. (Shortening occurs as the prestress is transferred to concrete)

Loss of Prestress

- In post-tensioned member,

  - if all tendons are to be stressed at the same time, there is no loss due to elastic shortening since the prestressing force is monitored during stressing process (i.e. after elastic shortening has occurred)

  - if the tendons are to be stressed sequentially, then there is no loss in the last tendon but maximum loss in the first tendon. The average loss of each tendon will then be one-half of the loss as if each tendon is having a maximum prestress loss.

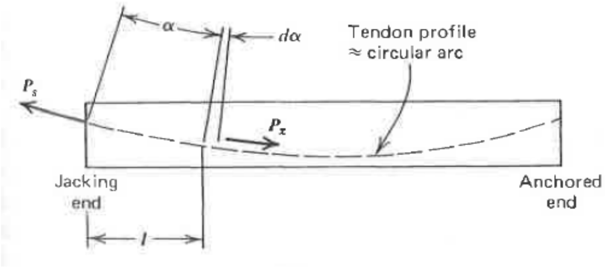
Loss of Prestress

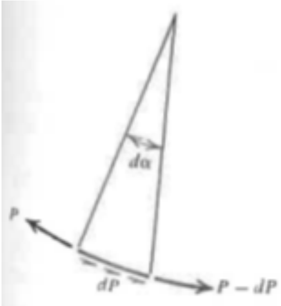
- Friction – For straight tendon, there is no friction loss due to stressing as there is no force induced to concrete.

- However, for linear tendon with direction change (kink), there will be vertical force acting on the concrete at the kink.

- During stressing process, this normal force will create friction (related by the coefficient of friction) at the kink.

- Similarly, for curved tendon, friction will act on the tendon along the entire length of the tendon, as the curvature of the tendon is changing continuously. *P* decreases continuously along tendon.





Loss of Prestress

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- Anchorage draw-in – For post-tensioning, after tensioning of tendon, the jack is released and the prestress is transferred to the anchorage. The anchorage fixtures that are subjected to stresses at this transfer will tend to deform, thus allowing the tendon to slacken slightly.

- Friction wedges employed to hold the wires will slip a little distance before the wires can be firmly gripped. The amount of slippage depends on the type of wedge and the stress in the wires (3 – 6 mm).

Loss of Prestress

<u>Long-term loss</u>

- The loss is also known as time-dependent loss (i.e. prestressing force varies with time)
- Creep of concrete – Additional strain under sustained loading. The additional compressive strain in concrete will shorten the tendon and reduce the prestressing force.
- Shrinkage of concrete – Withdrawal of water (unreacted/surplus) from concrete stored in unsaturated air causes drying shrinkage. This reduces the volume of concrete and reduces the force in tendon.
- Steel relaxation – This is the reduction in the prestressing force in wires under constant elongation maintained over a long period of time.

Safety

- High stresses exist in prestressed concrete members at BOTH maximum and minimum loading conditions.

- Limiting tendon zone exists – to satisfy allowable design stresses at top and bottom concrete fibres, which leads to design complication.

- Large jacking forces are required in the prestressing process. Careful design of anchorage block and full protection should be provided to the workers in case the tendons fail unexpectedly.

- Prestressed concrete structures are stored with a lot of energies. This makes the demolition of prestressed concrete structures different from ordinary reinforced concrete structures.

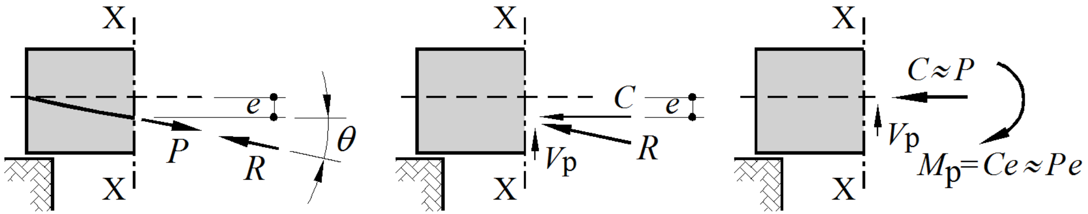
Summary

- What is prestressing concrete? And why?
- Advantages and Disadvantages
- Methods of prestressing: Pre- and post-tensioning; Internal and External prestressing
- Stress evaluation of members: At transfer (minimum load); At service (maximum load)
- Design implications
- Line of pressure (line of action)
- Straight, deflected and curved tendons
- Loss of prestress

Flexure Behaviour uncracked prestressed beams

Flexural Behaviour of uncracked beams

(a) uncracked beam with curved cable profile



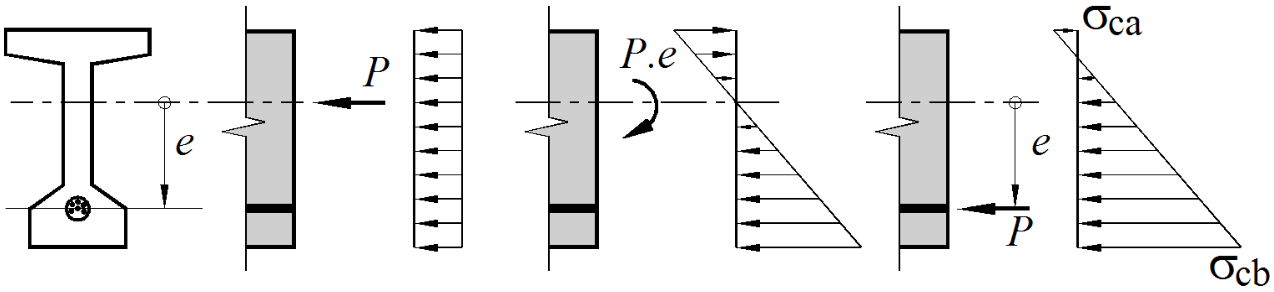
(b) forces in cable and concrete **(c) forces components in concrete** **(d) stress resultants in concrete**

Stress resultants in concrete due to prestress

With P=C. the concrete compressive stress at any depth y below the centroidal axis given by

$$\sigma_{cy} = \frac{P}{A_g} + \frac{Pe}{I_g} y$$

Stresses in a beam section due to prestress – ignoring all external loads and self weight



(a) Stresses due to axial force *P*

(b) Stresses due to *Pe*

(c) Stresses due to eccentric prestress

Stress at the bottom fibre of the beam after prestress

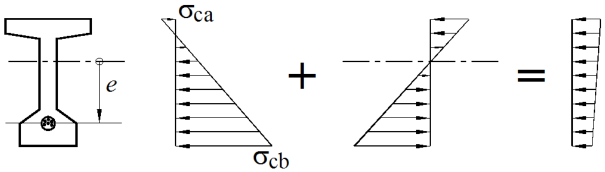
$$\sigma_{cb} = \frac{P}{A_g} + \frac{Pe}{I_g} y_b$$

Stress at the top fibre of the beam after prestress

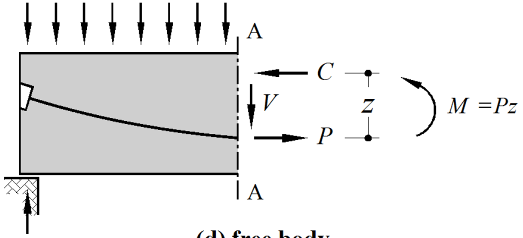
$$\sigma_{ca} = \frac{P}{A_g} - \frac{Pe}{I_g} y_t$$

P= prestress, *A* = cross sectional area, *e*= Eccentricity of prestress, ight, *I*= second moment of area, y_b and y_t distance from NA to the top or bottom edge of the beam

Stresses in a beam section due to prestress plus applied loads (applied moment)



(a) Prestress (b) Applied Moments (c) Resultant stresses



(d) free body

Positive moment M in section due to applied loads produces tensile stress in lower fibres of concrete. Elastic concrete stress at depth y below centroidal axis is given by $\sigma = -My/I_g$ *tension taken as negative*

When M is applied, <u>resultant</u> compressive stress due to prestress is reduced

Resultant compressive C force in concrete acts at distance z above "line of action" of the tensile force P

Horizontal forces in section, P and C are equal and comprise a couple which is equal to applied M:

$$M = Cz = Pz$$

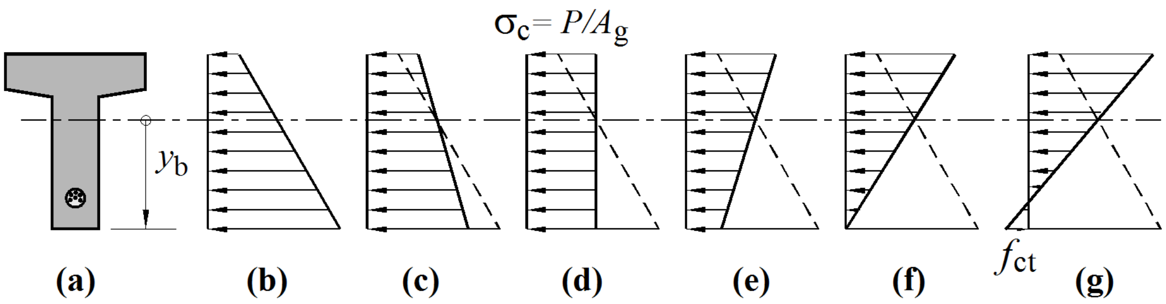
As M increases, P an C remain constant, "line of action" of C moves up in section, and z increases as M increases

Stresses in a beam section due to prestress plus applied loads (applied moment)

C and *P* remain constant, increase in applied *M* accommodate by increase in *z*

As applied moment *M* increases, the concrete compressive stress block changes shape

Concrete stress at level of the centroid does not change $\sigma_c = P/A_g$ (Stress line rotates about centroid)



b) $M = 0$

d) Uniform compressive stress $M = M_p = Pe$ (balanced)

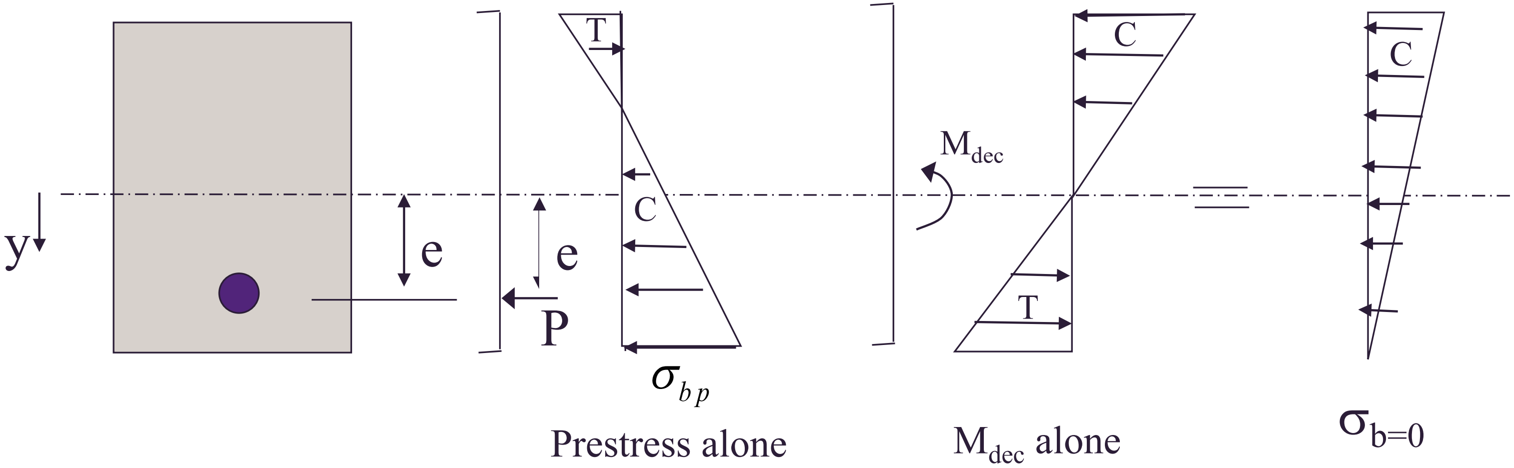
f) Stress is bottom fibre equal to zero, called decompression

g) Final stage reaches concrete tensile strength f_{ct} (cracked section analysis week 3)

Decompression moment

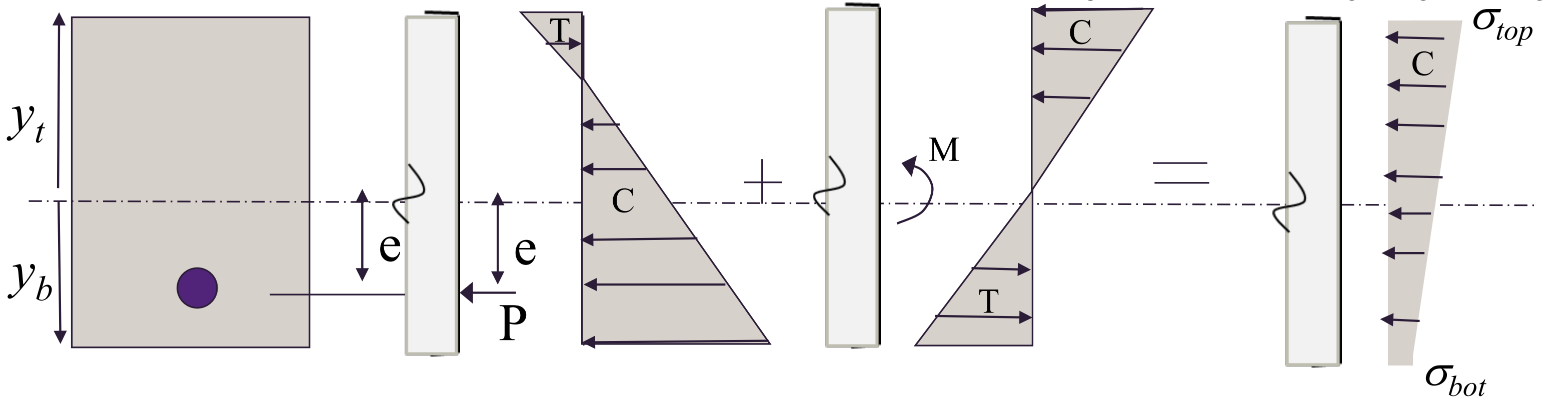
Applied moment at which the bottom fibre stress of the beam becomes zero

$$M_{dec} = \frac{\sigma_{cbp} I_g}{y_b}$$



Stresses in a beam section due to prestress and applied moment

$$\sigma_{ca} = \frac{P}{A_g} - \frac{Pe}{I_g} y_t \qquad \sigma_{top} = \frac{M}{I_g} y_t \qquad \sigma_{bot} = \frac{P}{A_g} + \frac{Pe}{I_g} y_t - \frac{My_t}{I_g}$$



$$\sigma_{cb} = \frac{P}{A_g} + \frac{Pe}{I_g} y_b \qquad \sigma_b = -\frac{M}{I_g} y_b \qquad = \qquad \sigma_{bot} = \frac{P}{A_g} + \frac{Pe}{I_g} y_b - \frac{My_b}{I_g}$$

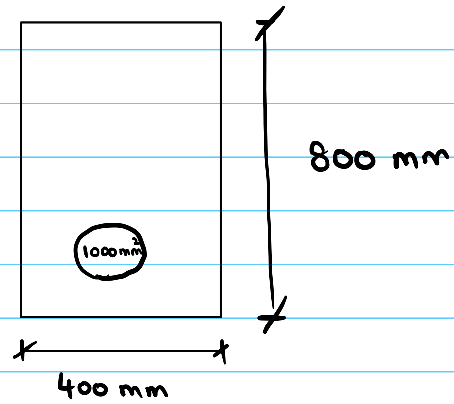
Stresses due to eccentric pretress + Applied moment = Combined stresses

Example of calculation of stresses

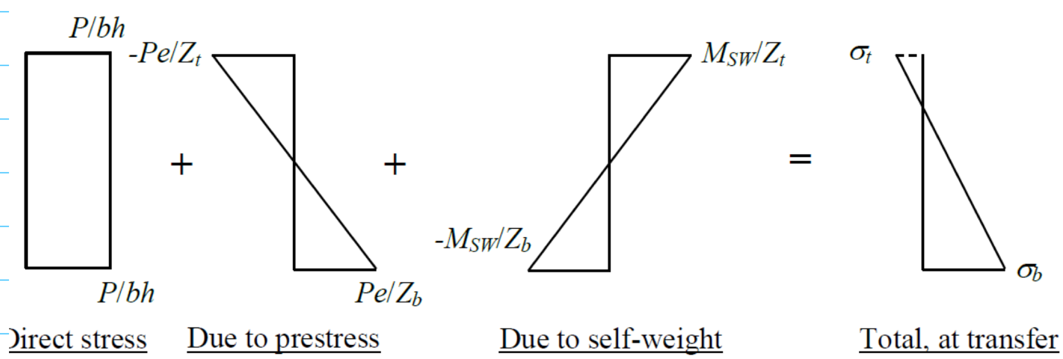
A prestressed concrete beam spans 10 m and has a rectangular cross-section 800 mm deep and 400 mm wide. It is post-tensioned by a cable having a cross-sectional area of 1000 mm2. The cable eccentricity varies parabolically from zero at each end to a maximum of 250 mm at midspan, where the prestressing force is 1200 kN. f'c = 32 MPa

Determine

1. The stresses at mid-span due to prestress acting alone
2. The decompression moment
3. The stress at midspan due to prestress, self weight and a 30 kN/M load



$e = 250$ mm
$P = 1200$ kN
$f'_c = 32$ MPa



$$Z_t = \frac{I}{y_t} = \frac{1}{6}bh^2$$
$$Z_b = \frac{I}{y_b} = \frac{1}{6}bh^2$$

$A_g = 400 \times 800 = 320{,}000 \text{ mm}^2$

$Z_t = Z_b = \frac{1}{6}bh^2 = \frac{1}{6} \times 400 \times 800^2$
$= 42.67 \times 10^6 \text{ mm}^3$

Specific weight $= 25 \text{ kN/m}^3$
$\rho_{CONCRETE} = 2400 \text{ kg/m}^3$

SELF WEIGHT LOAD $= 25 \text{ kN/m}^3 \times 0.32 \text{ m}^2$

END OF LECTURE 1